B O C O N H TH NG B M N NG M T TR I

NGHI N C U THU T TO N I U KHI N I U H NG T M PIN NG D NG TRONG H TH NG PIN N NG L NG M T TR I

QUY N 3 L P TR NH V HI U CHU N

N 2 (m h nh 2 tr c) Quy n 3/3 tu n b o c o 5/10 10/10/2026

Ph ng ph p hybrid: thi n v n nh v th theo gi DS1307, LDR tinh ch nh theo ng ng, gi v tr theo l ch khi m y m . Ph n c ng: ESP32 DevKit, m ch ng l c c u H b n TIP41C, motor g t n c tr c v t, LCD I2C 1602.

CH NG 1. T NG QUAN T I

1.1. L do ch n t i

V tr bi u ki n c a M€t Tr i d ch chuy n theo gi v• thay ,i theo m a. V i t*f*m pin l„p c nh, h ng ph p tuy n ch ph h…p v i tia tr c ti p trong m t s th i i m; sai l†ch g‡c l•m gi^m th•nh ph n b c x tr c ti p thu …c. H† b m n„ng hai tr c c‡ th thay ,i c^ ph ng v v• g‡c cao, l• m t ph ng n k‰ thuŠt kh^o s t kh^ n ng duy tr h ng pin g n v i h ng M€t Tr i.

C c m h nh d ng c^m bi n quang c n x‹ l Œ sai kh c gi a c c c^m bi n, nhi•u v• b‡ng che. V i b n LDR €t Ž b n g‡c, vi†c so s nh hai c€p ch•o gi•p x c nh ph a c‡ i†n trŽ nh‘ h n, t c ph a nhŠn nh s ng m nh h n. i’u khi n suy ra g‡c c n quay r“i ”c l i sau m•i chu k– gi•p c c*f*u c n ch nh m• kh ng ph^i quay li n t c.

’ t•i …c l a ch”n nh–m x y d ng thuŠt to n b m n„ng hai tr c d ng b n LDR v• hai ng c g t n c tr c v t cho ph ng v , nghi ng; tŠp trung v•o c ch lŠp ma trŠn i†n trŽ, suy ra g‡c c n quay, gi i h n h•nh tr nh v• chu k– ”c l i c^m bi n.

1.2. T ng quan v t i

C m b n LDR €t Ž b n g‡c m€t pin, m•i c^m bi n ch ch ra ngo•i m t g‡c b–ng nhau ph n bi†t h ng s ng theo hai ph ng. ESP32 t nh sai l†ch quang theo ma trŠn b n LDR, suy ra g‡c ch cho hai ng c g t n c tr c v t Ž c^ ph ng v v• nghi ng, h“i ti p g‡c b–ng bi n trŽ xoay; c ng t„c h•nh tr nh gi i h n chuy n ng. C c nghi n c u v’ tracker hai tr c …c d ng l•m t•i li†u tham kh^o, kh ng d ng k t qu^ c a ch•ng thay cho ph•p o tr n m h nh n•y [10, 11].

ThuŠt to n c a ’ t•i kh ng c n t nh g‡c €t thi n v n trong v˜ng i’u khi n ch nh: b i’u khi n ”c ma trŠn b n LDR, suy ra g‡c c n quay b–ng quan h† l …ng gi c, quay tr c ti p t i g‡c ch theo h“i ti p bi n trŽ v• nh k– ”c l i hi†u ch nh.

1.3. M€c ti•u nghi•n c,u

M c ti u t,ng qu t l• x y d ng thuŠt to n i’u khi n m h nh t*f*m pin hai tr c b–ng c ch so s nh i†n trŽ c a b n LDR, i’u khi n l n l …t h ng nghi ng v• h ng xoay, sau ‡ t ”c l i theo chu k– duy tr tr ng th i c n b–ng nh s ng.

- T‚ng h…p nguy n lŒ c c linh ki†n, ph ng ph p b m n„ng v• quy c x c nh h ng nghi ng, h ng xoay c a t*f*m pin.
- ’ xu*f*t c m b n LDR, ESP32, hai ng c g t n c tr c v t cho ph ng v v• nghi ng, hai bi n trŽ h“i ti p g‡c, RTC v• c ng t„c h•nh tr nh.
- X y d ng quy t„c ma trŠn b n LDR, x c nh ph a nhŠn s ng m nh h n, suy ra g‡c c n quay cho t™ng tr c v• ”c l i sau m•i chu k–.
- LŠp quy tr nh ki m tra h ng chuy n ng, ng ng c n b–ng, th i gian ”c l i v• so s nh n ng l …ng v i t*f*m pin c nh.

1.4. *f*i t„…ng v ph†m vi nghi•n c‚u

C*f*u h nh ’ xu*f*t d ng ESP32, b n LDR, hai ng c g t n c tr c v t (ph ng v v• nghi ng) ‡ng c„t qua relay/c u H, hai bi n trŽ h“i ti p g‡c, RTC v• c ng t„c h•nh tr nh cho hai tr c; photodiode …c xem x•t Ž m c so s nh. T”a l„p €t, k ch th c t*f*m pin, m -men c c*f*u v• gi i h n g‡c c n …c x c nh theo m h nh th c t tr c khi ch t o.

Ph m vi tŠp trung v•o ”c ma trŠn i†n trŽ LDR, suy ra g‡c l†ch hai h ng nghi ng/quay, i’u khi n hai c c*f*u ch*f*p h•nh quay t i g‡c ch v• bˆo v† h•nh tr nh.

1.5. K‡t quˆ d‰ ki‡n

K t quˆ d ki n g“m:

- M h nh kh i v• ph ng n ph n c ng hai tr c c‡ s “ k t n i, gi i h n chuy n ng v• c c b c hi†u chušn.
- ThuŠt to n ma trŠn i†n trŽ 4 LDR suy ra g‡c ch, i’u khi n motor ph ng v v• motor nghi ng quay t i g‡c ch, d™ng khi v•o v ng ch t v• t ”c l i theo chu k–.
- Ch ng tr nh i’u khi n ESP32 c ng quy tr nh ki m th‹ t™ng cˆm bi n, hai ng c g t n c v• c ng t„c h•nh tr nh.
- B ch ti u v• quy tr nh ghi d li†u so s nh c c ch b m sau khi ho•n t*f*t m h nh.

y l• c c sˆn phšm d ki n; k t quˆ nh l …ng v’ sai s ho€c n ng l …ng ch …c b, sung sau khi m ph‘ng/ o ki m tr n c*f*u h nh › x c nhŠn.

1.6. Vai trŠ c‹a quyŒn l•p tržnh

Quy n n•y tr nh b•y ph n m'm c a h† hai tr c: m i tr ng Arduino IDE v• c ch n p cho ESP32, module ESP32 DevKit v i c c nh‡m l†nh, thi t k ch t o hi†u chušn m ch v• c kh , c c ph ng ph p ”c ADC, i'u khi n hai motor qua c u H, ”c DS1307, hi n th LCD I2C, ”c bi n trŽ hai kh p, c ng l u “ t™ng ph n v• l u “ t,ng qu t kœm code m•u hai tr c ho•n ch nh.

CH NG 2. C S• L• THUY‘T, T NG QUAN LINH KI N V M'I TR NG L P TR NH

2.1. Pin n“ng l„…ng M”t Tr•i

M - un quang i†n …c cfu th•nh t™ nhi'u t b•o b n d•n gh•p n i ti p/song song; m•i t b•o bi n ,i m t ph n b c x th•nh d˜ng i†n m t chi'u. M - un th ng c‡ k nh m€t tr c, l p ‡ng g‡i, c c t b•o, l p m€t sau/ k nh sau, khung v• h p n i; cfu t o c th t y lo i. Trong chu•i t b•o c‡ th c‡ diode bypass h n ch ˆnh h Žng c a che b‡ng c c b ; diode n•y kh c v i photodiode d ng l•m cˆm bi n nh s ng.

C c th ng s th ng d ng m tˆ pin g“m Pmax, Voc, Isc, Vmp, Imp v• h† s nhi†t. Gi tr danh nh …c c ng b theo i'u ki†n STC; ngo•i th c t , b c x , nhi†t t b•o, ph, nh s ng, tˆi i†n v• b‡ng che l•m €c tuy n I-V thay ,i. H ng €t ˆnh h Žng th•nh ph n b c x tr c ti p; kh ng th suy ra c ng suft th c ch t™ g‡c h nh h”c ho€c i†n p hŽ m ch [3].

Trong ph•p th‹ tracker, o U v• l “ng th i, t nh P = UžI v• t ch ph n theo th i gian thu n ng l …ng. Tfm i ch ng n n c ng lo i, c ng t nh tr ng b' m€t v• c ng tˆi; n ng l …ng ti u th c a hai ng c g t n c, ESP32 v• m ch ph tr… phˆi …c ghi ri ng r“i tr™ khi t nh n ng l …ng thu n.

2.2. Cˆm bi‡n quang tr– (LDR)

Quang trŽ (Light Dependent Resistor - LDR) l• linh ki†n quang d•n c‡ i†n trŽ thay ,i theo nh s ng. Khi r”i t ng, i†n trŽ LDR th ng giˆm; trong t i i†n trŽ l n h n. €c tuy n n•y kh ng tuy n t nh, thay ,i theo ph, ngu“n s ng, nhi†t , l sˆn xuft v• l ch s‹ chi u s ng; t c p ng th ng chŠm h n diode quang.

Trong m ch o, LDR th ng m„c v i i†n trŽ c nh th•nh c u ph n p; ADC ”c i†n p t i i m gi a vi i'u khi n x‹ lŒ. Chi'u t ng/giˆm c a s ADC khi s ng ph thu c v tr LDR trong c u ph n p. Ÿu i m c a LDR l• r , d• t m v• m ch n giˆn; h n

ch l• sai kh c gi a m•u, tr i theo m i tr ng v• tuy n t nh th*f*p, v vŠy c n bˆo v† kh‘i b i/šm v• hi†u chušn n u d ng cho ph•p o nh l …ng.

2.3. B— i u khiŒn ESP32 DevKit v c˜c kh*f*i ph€ tr…

B i’u khi n trung t m l• module ESP32 DevKit duy nh*f*t cho cˆ hai tr c. Bo c‡ hai nh‡m ADC 12 bit; ’ t•i kh ng bŠt WiFi n n d ng …c cˆ nh‡m ADC2. Theo sheet nguy n lŒ: b n k nh LDR v•o GPIO25/26/27/14, bi n trŽ h“i ti p g‡c ph ng v v•o GPIO36, i†n p t*f*m pin qua c u chia v•o GPIO39, bi n trŽ h“i ti p g‡c nghi ng c a bˆn hai tr c d ng GPIO2; b n c ng t„c h•nh tr nh qt1 qt4 v•o GPIO34/35/32/33. ADC c a ESP32 ch o i†n p 0 3,3 V, phi n bˆn n•y ch a o d˜ng i†n.

Th i gian th c d ng module DS1307 (I¡C 0x68, pin nu i) c*f*p ng•y gi cho kh i thi n v n; hi n th d ng LCD 1602 kœm m ch chuy n I¡C PCF8574 (0x27) chung bus SDA/SCL (GPIO21/22). Ngu“n 12 V qua diode ch ng ng …c c c 1N4007, LM2596 h 3,3 V cho ESP32 v• nh nh 7805 ri ng c*f*p 5 V cho LCD, DS1307.

2.4. V™ trš M”t Tr•i v g›c theo ng y, gi•

2.4.1. Hœ t a —v c˜c g›c ”c tr„ng

V tr M€t Tr i th ng …c m tˆ b—ng g‡c cao ¢ (t nh t™ ng ch n tr i l n thi n nh) v• g‡c ph ng v £ (o t™ h ng B„c thŠt theo chi’u kim “ng h“). G‡c thi n nh ¤z = 90¥ ¦ ¢. G‡c ph ng v v• g‡c cao l• u v•o x c nh h ng ph p tuy n c a t*f*m pin; khi c•i €t phˆi th ng nh*f*t quy c d*f*u gi a ph n m’m v• chi’u quay th c t c a c c*f*u.

2.4.2. •nh h„—ng c‹a ng y trong n“m v gi• trong ng y

M t x*f*p x th ng d ng cho g‡c l†ch M€t Tr i l• § ¨ 23,45¥ ž sin[360¥ ž (284 + n) / 365], trong ‡ n l• th t ng•y trong n m. G‡c gi x*f*p x © = 15¥ ž (t_solar ¦ 12), v i t_solar l• gi M€t Tr i a ph ng t nh b—ng gi . T™ vª «, c‡ th t nh g‡c cao b—ng quan h† sin ¢ = sin « sin § + cos « cos § cos ©. C c bi u th c n•y th ch h…p tr nh b•y nguy n lŒ; tri n khai ch nh x c n n d ng thuŠt to n v tr M€t Tr i SPA ho€c th vi†n › ki m ch ng [1, 2].

Gi d n d ng kh ng ho•n to•n tr ng v i gi M€t Tr i. Ch ng tr nh c n hi†u ch nh m•i gi , kinh a i m v• ph ng tr nh th i gian; “ng h“ RTC phˆi …c €t •ng. Sai l†ch th i gian,

nhŠp nh m vª /kinh ho€c x c nh sai h ng B„c thŠt ’u l•m xu*f* t hi†n sai s g‡c d thuŠt to n t nh to n •ng.

2.4.3. G›c ”t c‹a tracker hai tr€c

V i tracker ph ng v - g‡c cao, tr c ph ng v h ng ph p tuy n v’ g n£ v• tr c cao €t g n¢; n u d ng g‡c nghi ng so v i ph ng ngang th ¬ = 90¥ ¦ ¢. G‡c €t th c t b–ng g‡c t nh to n c ng offset l„p €t, sau ‡ …c gi i h n theo h•nh tr nh. Quy c ph ng v phˆi th ng nh*f* t (B„c thŠt, chi’u kim “ng h“ hay quy c kh c); g‡c i qua 0¥ c n x‹ lŒ sao cho c c*f* u ch”n ng quay ng„n v• kh ng xo„n d y.

- m h nh hai tr c, ph•p t nh h nh h”c cho h ng tham chi u, c˜n gi i h n c kh , v tr home, sai s g l„p v• h•nh tr nh motor quy t nh g‡c l†nh th c t . Khi M€t Tr i Ž d i ng ng g‡c cao vŠn h•nh, h† th ng d™ng ho€c v’ v tr ngh › ch”n.

2.5. C˜c mža trong n“m v quŸ †o biŒu ki‡n

Tr c quay c a Tr i *f* t nghi ng so v i m€t ph®ng qu‰ o, v vŠy x ch vª M€t Tr i thay ,i trong n m. - B„c b n c u, v•o kho^ng th ng 6 M€t Tr i l n cao v• th i gian ban ng•y d•i h n; v•o kho^ng th ng 12 M€t Tr i th*f* p h n v• ng•y ng„n h n. Quanh hai i m ph n th ng 3 v• th ng 9, d•i ng•y v• m g n b–ng nhau. y l• bi n thi n thi n v n, kh ng phˆi s thay ,i t ng t •ng v•o ng•y u m a kh t …ng.

Do qu‰ o thay ,i theo m a, gi M€t Tr i m”c/l€n v• ph ng v m”c/l€n c¯ng thay ,i. C ng m t t*f* m pin €t c nh c‡ th ph h…p h n v•o m t s m a; l…i ch c a b m n„ng kh ng gi ng nhau quanh n m. - c c vª mi’n B„c Vi†t Nam, v•o m a hœ M€t Tr i gi a tr a c‡ th Ž r*f* t cao v• l†ch v’ ph a B„c; m a ng ng i th*f* p h n v• ch y u Ž ph a Nam. M t tracker m t tr c ch quay trong m t m€t ph®ng n n c˜n sai l†ch theo ph ng vu ng g‡c v i tr c; tracker hai tr c kh„c ph c t t h n v’ h nh h”c nh ng t ng ph c t p v• n ng l …ng truy’n ng.

V vŠy, thi t k c n xem x•t cˆ m a, g‡c nghi ng ban u, b‡ng che v• th i ti t a ph ng. Kh ng n n suy ra m c t ng sˆn l …ng cˆ n m t™ m t ng•y n„ng ho€c m t l n o; c n so s nh trong c c kho^ng th i gian d•i v• ghi nhŠn i’u ki†n o [3, 4].

Giai o†n thi•n v“n – B c b˜n c¡u	Xšch v¢ M”t Tr•i (x£p x¤)	”c iŒm „•ng i biŒu ki‡n
Xu n: i m ph n th ng 3 n ch th ng 6	0¥ ° +23,44¥	cao M€t Tr i gi a tr a t ng; ng•y d•i d n.
H :ch th ng 6 n i m ph n th ng 9	+23,44¥ ° 0¥	M€t Tr i i cao; ng•y d•i r“i giˆm d n.
Thu: i m ph n th ng 9 n ch th ng 12	0¥ ° ¦23,44¥	Qu‰ o h th*f*p; ng•y ng„n d n.
ng: ch th ng 12 n i m ph n th ng 3	¦23,44¥ ° 0¥	M€t Tr i i th*f*p; ng•y ng„n r“i d•i d n.

M c ng•y ph n v• ch thay ,i nh± theo t™ng n m; m a kh t …ng t i t™ng a ph ng c¯ng kh ng ho•n to•n tr ng v i m a thi n v n. Bˆng tr n m tˆ xu h ng qu‰ o, kh ng thay cho s li†u th i ti t t i v tr th‹ nghi†m.

2.6. Ba ph„¥ng ph˜p b˜m n ng xem x¦t trong t i

2.6.1. Nh›m 1 VŠng h– theo th•i gian (thu•t to˜n thi•n v“n)

T nh v tr M€t Tr i t™ t”a l„p €t, ng•y v• gi ”c t™ DS1307, kh ng c n cˆm bi n s ng. X ch v theo ng•y trong n m:

§ = 23,45¥ ² sin[360¥²(284 + n)/365]

G‡c gi theo gi M€t Tr i t:

H = 15¥ ² (t ¦ 12)

G‡c cao M€t Tr i ¢ theo v « :

sin ¢ = sin « ² sin § + cos « ² cos § ² cos H

G‡c ph ng v £ t nh ti p t™ ¢, §, «. C€p (¢, £) ,i tr c ti p ra g‡c €t c a tr c nghi ng v• tr c ph ng v .

- Ÿu i m: ,n nh, kh ng ph thu c th i ti t, kh ng dao ng.
- Nh …c i m: c n c•i •ng t”a v• gi ; kh ng t s‹a sai s l„p €t v• r kh p 3D.

2.6.2. Nh›m 2 VŠng kšn džng cˆm bi‡n quang tr– (LDR)

B n LDR Ž b n g‡c t*f*m pin, m•i cˆm bi n g nghi ng ra ngo•i t m m t g‡c ¬_s = 30¥, kh ng d ng v ch ng n. Sai l†ch i’u khi n

c a hai tr c:

e1 = ADC(LDR_tr i) ¦ ADC(LDR_ph^i); e2 = ADC(LDR_tr n) ¦ ADC(LDR_d i)

- N u |e1| > ng ng: quay motor ph ng v theo d*f*u e1; n u |e2| > ng ng: quay motor nghi ng theo d*f*u e2.
- Ng …c l i th d™ng v ng ch t ch ng dao ng quanh v tr c n b–ng.
- Ÿu i m: n gi^n, r , ph^n ng theo s ng th c t , t s‹a sai s g l„p.
- Nh …c i m: nh y m y v• b‡ng r m tho ng qua; s ng s m t n hi†u y u n n b m chŠm.

2.6.3. Nh›m 3 Ph„¥ng ph˜p hybrid (k‡t h…p), ph„¥ng ˜n l‰a ch n

- Thi n v n nh v th : u bu,i s ng v• m•i 30 ph•t a nhanh c^ hai tr c v' g n (¢, £) c a M€t Tr i.
- LDR tinh ch nh: gi a hai l n nh v th , d ng e1 ch nh ph ng v v• e2 ch nh nghi ng v' v ng ch t.
- Tr i nhi'u m y (t,ng s ng S < S_min): gi nguy n v tr theo l ch thi n v n, tr nh dao ng v ch.

Hybrid gi u i m c a c^ hai nh‡m v• kh„c ph c nh …c i m c a t™ng nh‡m, n n …c ch"n cho m h nh hai tr c c a " n n•y.

2.7. Nguy•n t c i u khiŒn v ˜nh gi˜

C‡ th d ng i'u khi n ‡ng-ng„t c‡ v ng tr• t m c ti u nhanh; i'u khi n t³ l† gi^m b c khi g n ch; ho€c PI/PID khi c‡ ph^n h"i v tr v• c n gi^m sai s ,n nh. V i motor tr c v t, l†nh quay …c gi i h n theo h•nh tr nh v• ch ph t khi g‡c l†ch v …t v ng ch t; nh tr c v t t h›m n n kh ng c n nu i i†n gi v tr . ThuŠt to n m ho€c th ch nghi l• h ng n ng cao nh ng c n …c hi†u chušn v• so s nh tr n d li†u i di†n [4].

Ch ti u nh gi n n g"m sai s g‡c ph ng v /g‡c cao, s l n chuy n ng, th i gian ,n nh, tr ng th i c ng t„c v• n ng l …ng thu n. N ng l …ng PV v• n ng l …ng ng c /b i'u khi n c n o tr n c ng kho^ng th i gian; trong c c ph•p so s nh ph^i gi nguy n t^i, c*f*u h nh o v• i'u ki†n l„p €t.

2.8. M§i tr„•ng Arduino IDE v c˜ch n†p ch„¥ng trŽnh cho ESP32

Arduino IDE m€c nh ch kœm l´i AVR/SAM v• tr nh bi n d ch avr-gcc/arm-gcc, trong khi ESP32 d ng nh n Xtensa LX6 c n toolchain ri ng, n n phˆi c•i th m l´i esp32 c a Espressif th IDE m i bi n d ch v• n p …c cho ESP32 DevKit.

- B c 1. C•i Arduino IDE v• driver USB-UART (CP210x/CH340).
- B c 2. File ° Preferences ° Additional Boards Manager URLs: th m https://espressif.github.io/arduino-esp32/package_esp32_index.json
- B c 3. Tools ° Board ° Boards Manager ° c•i g‡i µesp32¶ c a Espressif Systems.
- B c 4. Ch"n Board µESP32 Dev Module¶ v• •ng c,ng COM.
- B c 5. Serial Monitor 115200 baud; gi n•t BOOT khi n p n u IDE b o µConnecting...¶.
- B c 6. N p sketch Blink x c nhŠn tr c khi vi t ch ng tr nh hybrid hai tr c.

2.9. Module ESP32 DevKit džng trong t i

Module ESP32 DevKit i'u khi n cˆ hai kh p. ' t•i kh ng bŠt WiFi n n d ng …c nh‡m ADC2 cho b n k nh LDR (GPIO25/26/27/14); bi n trž ph ng v v•o GPIO36, bi n trž nghi ng v•o GPIO2, i†n p tƒm pin v•o GPIO39 (ADC1). L†nh hai c u H i qua b n opto PC817 v i quy c t ch c c thƒp; b n c ng t„c h•nh tr nh k•o xu ng 1k "c m c 1 khi ch m.

2.10. C˜c nh›m lœnh c¥ bˆn džng trong ch„¥ng tržnh

Bˆng 1. C c nh‡m l†nh d ng trong ch ng tr nh hybrid hai tr c

Nh›m	Lœnh †i diœn	ng d€ng trong t i
Cƒu tr•c	if/else, switch-case, for, while	Quy t nh l†nh theo dƒu e1/e2; v˜ng ch hai motor.
V•o/ra s	pinMode, digitalWrite, digitalRead	B n k nh l†nh c u H; b n c ng t„c h•nh tr nh.
V•o/ra t ng t	analogRead, analogSetPinAttenuation	B n k nh LDR (ADC2), hai bi n trž v• p tƒm pin (ADC1).
Th i gian	millis, delay	Chu k– 2 ph•t nh nh LDR, 30 ph•t nh nh thi n v n.

Nh›m	Lœnh †i diœn	ng d€ng trong t i
To n h”c	sin, cos, atan2, radians, degrees	T nh §, H, ¢, £ v• ,i ra g‡c €t hai kh p.
I¡C	Wire.beginTransmission(0 x68/0x27)	”c DS1307, ghi l†nh LCD PCF8574.
Hi n th	lcd.setCursor, lcd.print	Gi , hai g‡c kh p, tr ng th i hai motor.
G l•i	Serial.printf	Log b n k nh, e1, e2, hai g‡c €t.

2.11. Th„ viœn s¨ d€ng

- Wire.h cho DS1307 (0x68) v• PCF8574 c a LCD (0x27) tr n c ng bus I¡C.
- LiquidCrystal_I2C.h i’u khi n LCD 1602 qua hai ch n SDA/SCL.
- To n h”c chušn C++ cho kh i thi n v n; kh ng d ng WiFi/Bluetooth gi nh‡m ADC2.

2.12. C˜c tham sƒ v k‡t qu^ k‡ th©a t© quyŒn nghi•n c„u

B^ng 2. Tham s k th™a t™ quy n nghi n c u a v•o ch ng tr nh hai tr c

†i l„…ng	Gi˜ tr™ ª thƒng nh£t trong quyŒn nghi•n c„u	Džng trong code
Chu k– thi n v n / LDR	30 ph•t / 2 ph•t	T_TV = 1800000 ms; T_LDR = 120000 ms.
Ng ng / v ng ch t	|e| > 200 / 120 m c ADC	H–ng s NGUONG v• i’u ki†n d™ng.
Ng ng m y m S_min	2500 (t„ng b n k nh)	Nh nh gi v tr c^ hai kh p.
Th t tinh ch nh	Kh p nghi ng (e2) tr c, ph ng v (e1) sau	Th t g”i trong nh nh LDR c a loop().
G‡c g c^m bi n ¬_s	30¥, ch nghi ng quang trŽ	C sŽ hi†u chušn K_cal b n k nh.
Sai s luŠt hybrid	2,9¥ trung b nh; 83 l n ch y motor/ng•y	M c so s nh log o ngo•i tr i.
Ma trŠn (¢, £) v• chušn h‡a 180¥	Ch ng 4 quy n nghi n c u	H•m thienVan() v• b^ng i chi u.

CH NG 3. CH‘ T«O, HI U CHU N, O «C V C C PH NG PH P X¬ L• TRONG CODE

3.1. Thi‡t k‡ m†ch tr•n EasyEDA

M ch …c v· tr n EasyEDA th•nh bˆy sheet nh bˆn m t tr c, ri ng bˆn hai tr c d ng cˆ hai m ng l†nh motor (th_thuan/th_nguoc v• th_thuan2/th_nguoc2) v• th m bi n trŽ kh p nghi ng Ž GPIO2. C c quy t nh thi t k ˆnh h Žng t i code: b n k nh LDR tr n ADC2 n n kh ng bŠt WiFi; l†nh c u H t ch c c th*f*p qua opto; c ng t„c h•nh tr nh k•o xu ng 1k ”c m c 1 khi ch m; DS1307 v• LCD chung bus I¡C.

3.2. Ch‡ t†o m†ch

- Ho•n th•nh h•n kh i ngu“n ba nh nh v• o 3,3 V / 5 V tr c khi c„m module.
- Ho•n th•nh h•n hai c u H t m TIP41C kœm tˆn nhi†t, diode bˆo v† v• b n k nh opto c ch ly.
- Ho•n th•nh i d y bus I¡C v• d y t n hi†u b n k nh LDR xo„n i tr nh nhi•u c u H.

3.3. Thi‡t k‡ c¥ khš ph€c v€ ph¡n m m

- Hai bi n trŽ h“i ti p g„n “ng tr c t™ng kh p code n i suy tuy n t nh ba i m.
- B n c ng t„c h•nh tr nh €t ngo•i v ng l•m vi†c 2¥, code d ng l•m gi i h n c ng t™ng chi’u.
- Kh i g LDR in 3D gi nguy n g‡c nghi ng ¬_s sau hi†u chušn h† s K_cal ,n nh gi a c c l n ch y.

3.4. Hiœu chu-n v o †c

- Hi†u chušn K_cal b n k nh LDR khi c‡ m ch; sai l†ch sau hi†u chušn d i 2% l• gi tr ch tham khˆo, s· ki m ch ng l i.
- Hi†u chušn hai bi n trŽ h“i ti p ba i m; sai s g‡c suy ra ¸ 1,5¥ m•i kh p l• gi tr tham khˆo, s· ki m ch ng l i.
- o tr i gi DS1307 v• th‹ hi n th hai d˜ng LCD c ng vi†c ti p theo; m c tr i cho ph•p theo t•i li†u nh• sˆn xu*f*t l• tham khˆo.
- o d˜ng hai motor khi c‡ tˆi c ng vi†c ti p theo; gi tr tham khˆo theo t•i li†u ng c c 4 6 A khi tˆi, code gi i h n m•i l†nh 8 gi y.

3.5. Ph„¥ng ph˜p c ADC trong code

- S u k nh t ng t ”c 12 bit v i suy hao 11 dB; m•i k nh l*f*y 16 m•u, b‘ m•u l†ch r“i trung b nh.
- Nh n K_cal t™ng k nh LDR tr c khi lŠp e1 = tr i ¦ ph^i v• e2 = tr n ¦ d i; ng ng 200 v• v ng ch t 120 m c ADC.
- K nh p t*f*m pin qua c u chia ch gi m s t v• hi n th ; phi n b^n n•y ch a o d˜ng i†n.

3.6. Ph„¥ng ph˜p i u khiŒn hai —ng c¥ qua c¡u H

B^ng 3. B^ng l†nh c u H d ng chung cho hai kh p

T*Ž*nh hu*f*ng	Lœnh code	K‡t qu^
Kh p ch y thuŠn	thuan = LOW, nguoc = HIGH	C€p ch•o tr i d•n.
Kh p ch y ng …c	thuan = HIGH, nguoc = LOW	C€p ch•o ph^i d•n.
D™ng / gi	c^ hai = HIGH	C u H kh‡a, tr c v t t h›m.
Ch m h•nh tr nh	d™ng + c*f*m chi’u ch m	B^o v† c kh v• transistor.

Th t tinh ch nh trong m•i chu k– LDR l• kh p nghi ng tr c r“i kh p ph ng v sau, sai l†ch ph ng v …c o khi m€t pin › •ng cao; m•i l†nh ch y …c gi m s t 10 ms m t l n v• t kh‡a c u H khi t v ng ch t ho€c h t 8 gi y.

3.7. c th•i gian t© module DS1307

Code ”c b^y thanh ghi BCD c a DS1307 qua a ch 0x68, gi^i m› BCD r“i suy ra s ng•y trong n m n v• gi M€t Tr i t t nh §, H, ¢, £; hai g‡c €t c a kh p nghi ng v• kh p ph ng v l*f*y tr c ti p t™ c€p (¢, £) sau khi chušn h‡a b c nh^y 180¥ tr a h ch .

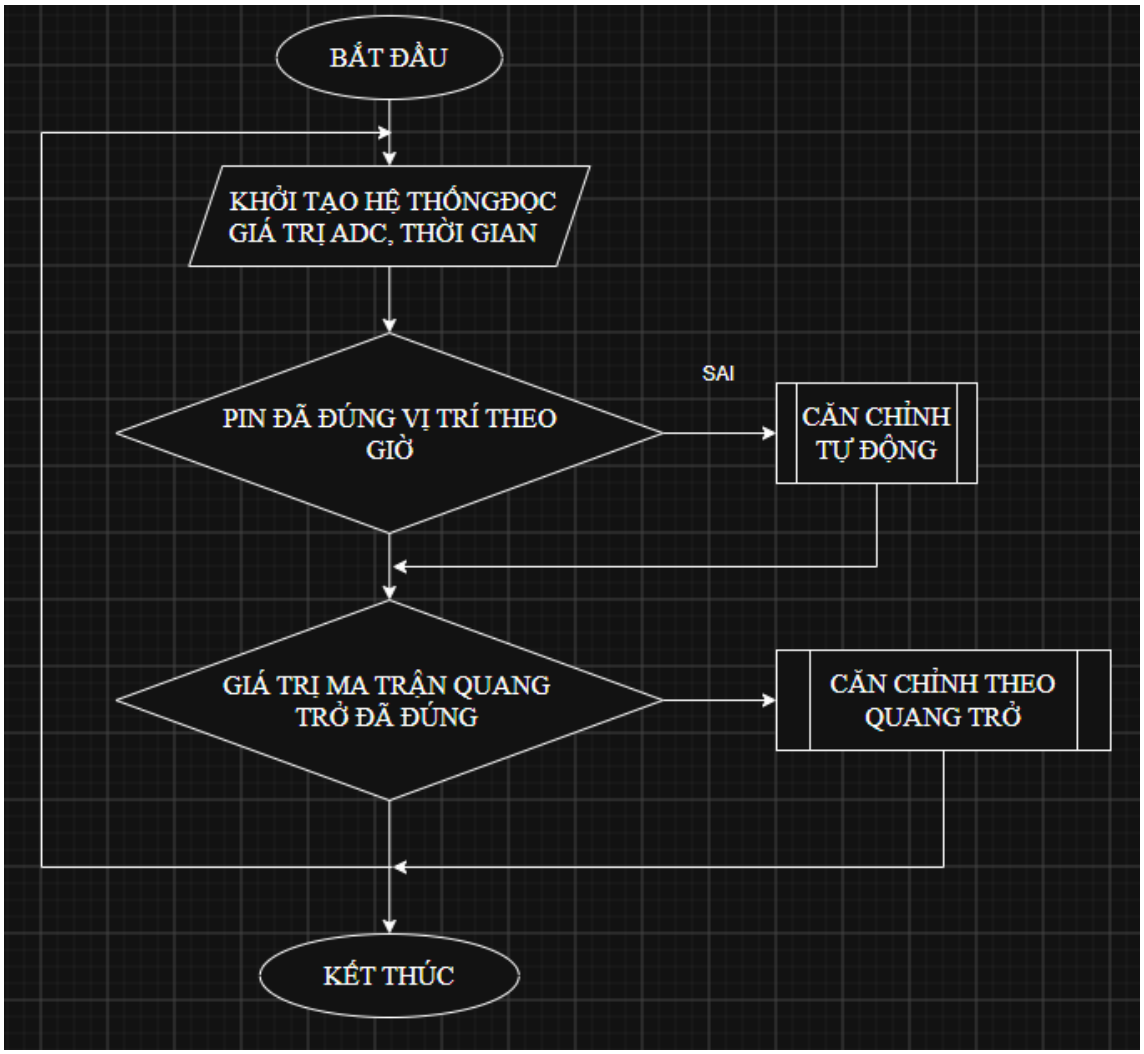
3.8. HiŒn th™ th•i gian v tr†ng th˜i l•n LCD I2C 1602

M•i gi y, d˜ng 1 LCD in ng•y gi t™ DS1307 d ng µDD/MM HH:MM:SS¶; d˜ng 2 in g‡c hai kh p v• tr ng th i, v d µng=+35 q=-12 OK¶. Khi m y m gi v tr ho€c ch m h•nh tr nh, d˜ng 2 ,i th•nh th ng b o µGIU VI TRI¶ ho€c µCHAN HANH TRINH¶.

3.9. c bi‡n tr– suy ra g›c nghi•ng v g›c ph„¥ng v™

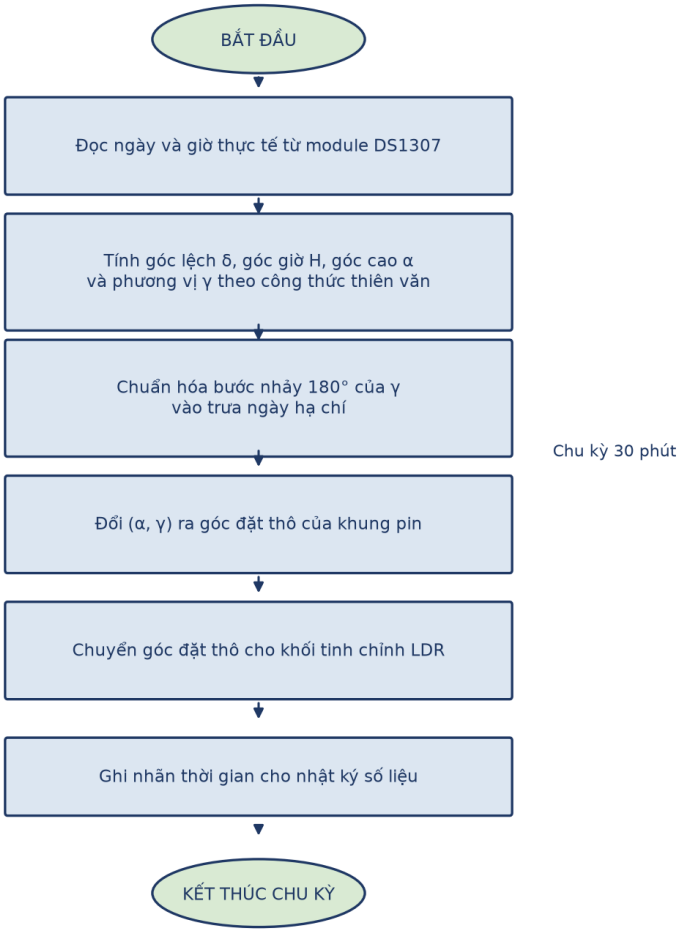
M•i kh p m t bi n trŽ chia p 3,3 V; code ”c trung b nh 16 m•u r“i n i suy ba i m hi†u chušn ra g‡c kh p, k±p trong h•nh tr nh c kh (nghi ng 0 75¥, ph ng v ¹120¥). Gi tr n•y d ng gi i h n g‡c €t thi n v n, d™ng l†nh tinh ch nh v• hi n th LCD.

3.10. L„u ® thu•t to˜n t©ng ph¡n v t ng qu˜t



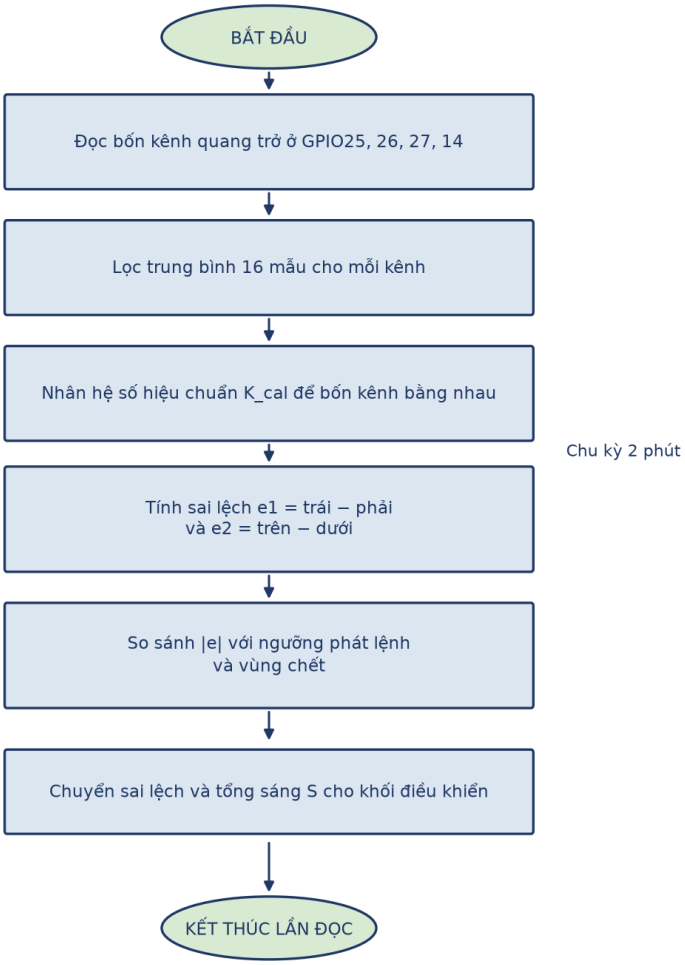
H nh 1. L u t ng qu t ch ng tr nh hybrid hai tr c (b n v c a sinh vi n)

L u “ t,ng qu t bˆn sinh vi n cho h† hai kh p: v˜ng ki m tra v tr theo gi ch y tr c, t i v˜ng tinh ch nh ma trŠn quang trŽ th x•t kh p nghi ng r“i m i n kh p ph ng v .



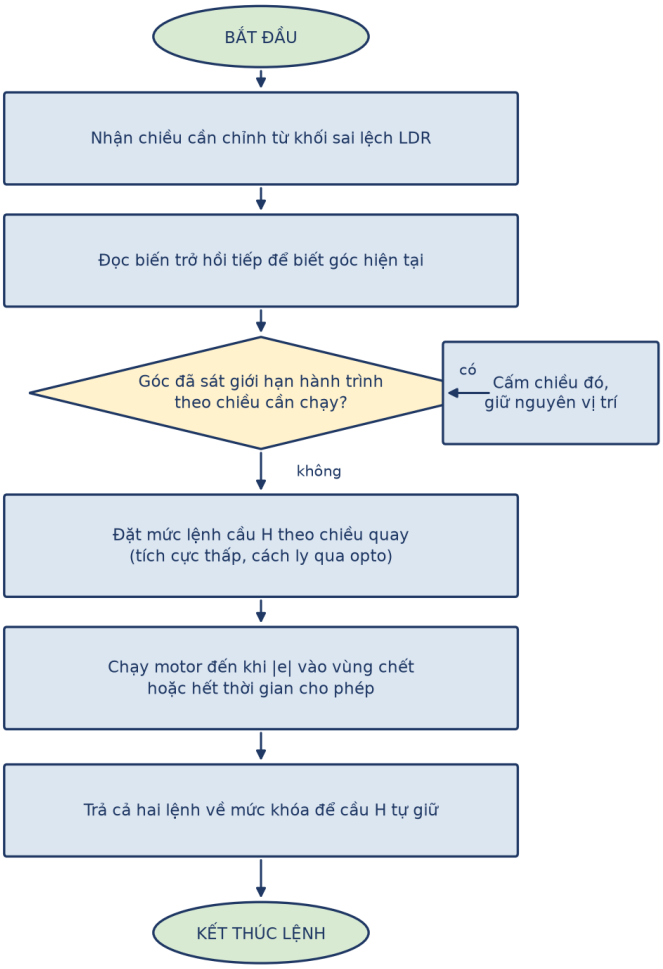
H nh 2. L u c gi DS1307 v t nh g c thi n v n

Kh i thi n v n hai kh p ”c DS1307 m•i 30 ph•t, t nh g‡c cao v• ph ng v r“i ,i ra g‡c €t nghi ng v• ph ng v , c‡ b c chušn h‡a nh^y 180¥ tr a h ch .



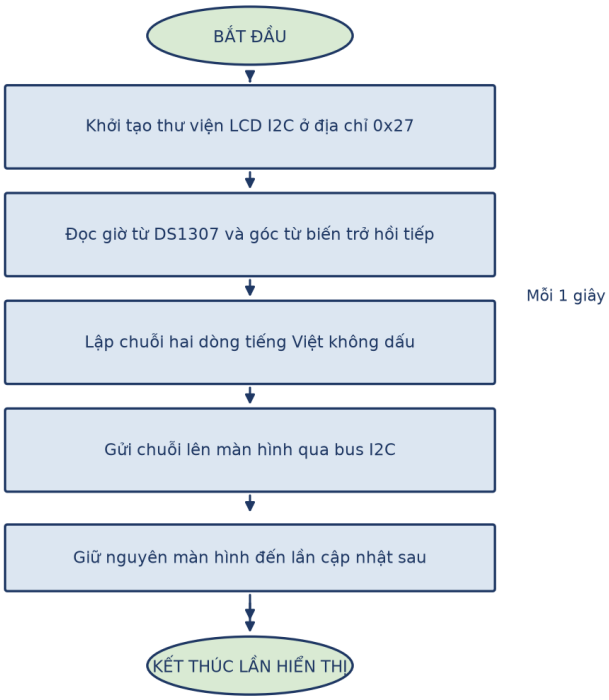
H nh 3. L u c b n k nh LDR v t nh e1, e2

Kh i ”c b n k nh LDR lŠp “ng th i sai l†ch e1 cho ph ng v v• e2 cho g‡c nghi ng, nh n h† s K_cal v• so ng ng tr c khi chuy n cho kh i i’u khi n hai kh p.



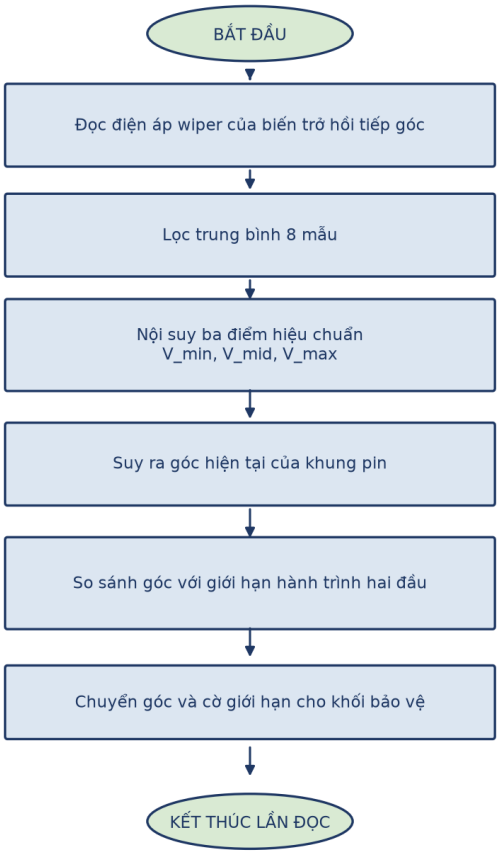
H nh 4. L u i u khi n motor qua c u H b n TIP41C

Kh i i'u khi n ch y l n l …t hai c u H: kh p nghi ng ch nh tr c, kh p ph ng v ch nh sau; m•i kh p 'u ki m tra gi i h n h•nh tr nh b–ng bi n trŽ ri ng tr c khi c*f*p l†nh.



H nh 5. L u hi n th LCD I2C 1602

Kh i hi n th a gi th c, g‡c nghi ng v• g‡c ph ng v l n hai d˜ng LCD I2C m•i gi y ng i vŠn h•nh theo d´i.



H nh 6. L u c bi n tr suy ra g c kh p

Kh i h“i ti p ”c hai bi n trŽ “ng tr c, n i suy ba i m hi†u chušn cho t™ng kh p v• b o c gi i h n h•nh tr nh v’ kh i bˆo v†.

3.11. Code m¯u ho n ch¤nh c‹a m§ hŽnh hai tr€c

```
// Bam nang hybrid 2 truc - ESP32 DevKit (thien van tho + LDR tinh chinh)
// Nghien truoc, phuong vi sau. Chan theo sheet: LDR 25/26/27/14,
// bien tro PV 36 / nghien 2, ap pin 39, hanh trinh 34/35/32/33,
// motor PV 19/18, motor nghien 5/13 (tich cuc THAP qua opto).
#include <Arduino.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
const int LDR[4] = {25, 26, 27, 14};       // TT, PT, TD, PD (ADC2)
const int POT_PV = 36, POT_NG = 2, PIN_AP = 39;
const int QT[4]  = {34, 35, 32, 33};       // hanh trinh: PV+, PV-, NG+, NG-
const int PV_T = 19, PV_N = 18;            // cau H motor phuong vi
const int NG_T = 5,  NG_N = 13;            // cau H motor nghien
LiquidCrystal_I2C lcd(0x27, 16, 2);
const float PHI = 20.93;
const int NGUONG = 200; float S_MIN = 2500;
const unsigned long T_LDR = 120000, T_TV = 1800000;
float K_cal[4] = {1.0, 1.0, 1.0, 1.0};
unsigned long tLDR = 0, tTV = 0;
int docTB(int chan, int n = 16) {
  long s = 0; int m[16];
  for (int i = 0; i < n; i++) { m[i] = analogRead(chan); s += m[i]; }
  long tb = s / n; long s2 = 0; int d = 0;
  for (int i = 0; i < n; i++) if (abs(m[i] - tb) * 4 < tb) { s2 += m[i]; d++; }
  return d ? s2 / d : tb;
}
void docDS1307(int &ng, int &th, int &hh, int &mm, int &ss) {
  Wire.beginTransmission(0x68); Wire.write(0x00); Wire.endTransmission();
  Wire.requestFrom(0x68, 7);
```

```
  byte b[7]; for (int i = 0; i < 7; i++) b[i] = Wire.read();
  auto bcd = [](byte x) { return (x >> 4) * 10 + (x & 0x0F); };
  ss = bcd(b[0] & 0x7F); mm = bcd(b[1] & 0x7F); hh = bcd(b[2] & 0x3F);
  ng = bcd(b[4] & 0x3F); th = bcd(b[5] & 0x1F);
}
int ngayTrongNam(int ng, int th) {
  const int t31[12] = {31,28,31,30,31,30,31,31,30,31,30,31};
  int n = ng; for (int m = 1; m < th; m++) n += t31[m - 1];
  return n;
}
// thien van: tra ve (alpha, gamma) do
void thienVan(float &alpha, float &gamma) {
  int ng, th, hh, mm, ss; docDS1307(ng, th, hh, mm, ss);
  int n = ngayTrongNam(ng, th);
  float t = hh + mm / 60.0 + ss / 3600.0;
  float d = radians(23.45 * sin(radians(360.0 * (284 + n) / 365.0)));
  float H = radians(15.0 * (t - 12.0));
  float p = radians(PHI);
  float sa = sin(p)*sin(d) + cos(p)*cos(d)*cos(H);
  if (sa <= 0.05) { alpha = 0; gamma = 0; return; }
  alpha = degrees(asin(sa));
  gamma = degrees(atan2(sin(H), cos(H)*sin(p) - tan(d)*cos(p)));
  if (gamma > 90.0) gamma -= 180.0;        // chuan hoa buoc nhay trua ha chi
  if (gamma < -90.0) gamma += 180.0;
}
float docGoc(int chan, float gMin, float gMax) {
  float v = docTB(chan) * 3.3 / 4095.0;
  const float VMIN = 0.30, VMAX = 3.00;
  return constrain(gMin + (v - VMIN) * (gMax - gMin) / (VMAX - VMIN), gMin, gMax);
}
void dung(int t, int n) { digitalWrite(t, HIGH); digitalWrite(n, HIGH); }
void chay(int t, int n, int qtThuan, int qtNguoc, bool thuan, int ms) {
  if (thuan && digitalRead(qtThuan) == 1) return;
  if (!thuan && digitalRead(qtNguoc) == 1) return;
  digitalWrite(t, thuan ? LOW : HIGH);
  digitalWrite(n, thuan ? HIGH : LOW);
  delay(ms); dung(t, n);                   // cau H khoa, truc vit tu giu
}
void setup() {
  Serial.begin(115200);
  Wire.begin(); lcd.init(); lcd.backlight();
  analogReadResolution(12);
  analogSetPinAttenuation(POT_PV, ADC_11db);
  analogSetPinAttenuation(POT_NG, ADC_11db);
  analogSetPinAttenuation(PIN_AP, ADC_11db);
  for (int i = 0; i < 4; i++) { analogSetPinAttenuation(LDR[i], ADC_11db); pinMode(QT[i], INPUT); }
  pinMode(PV_T, OUTPUT); pinMode(PV_N, OUTPUT); dung(PV_T, PV_N);
  pinMode(NG_T, OUTPUT); pinMode(NG_N, OUTPUT); dung(NG_T, NG_N);
}
void loop() {
  unsigned long now = millis();
  // 1) dinh vi tho ca hai khop moi 30 phut
  if (now - tTV >= T_TV) {
    tTV = now;
    float a, g; thienVan(a, g);
    float ngHien = docGoc(POT_NG, 0, 75), pvHien = docGoc(POT_PV, -120, 120);
    if (fabs(a - ngHien) > 2.0)
      chay(NG_T, NG_N, QT[2], QT[3], a > ngHien, min(5000, (int)fabs(a - ngHien) * 80));
    if (fabs(g - pvHien) > 2.0)
      chay(PV_T, PV_N, QT[0], QT[1], g > pvHien, min(5000, (int)fabs(g - pvHien) * 80));
  }
  // 2) tinh chinh LDR moi 2 phut: nghien truoc, phuong vi sau
  if (now - tLDR >= T_LDR) {
    tLDR = now;
    int L[4]; float S = 0;
    for (int i = 0; i < 4; i++) { L[i] = docTB(LDR[i]) * K_cal[i]; S += L[i]; }
    float e2 = (L[0] + L[1]) - (L[2] + L[3]);   // tren - duoi
    float e1 = (L[0] + L[2]) - (L[1] + L[3]);   // trai - phai
    if (S > S_MIN) {
      if (fabs(e2) > NGUONG) chay(NG_T, NG_N, QT[2], QT[3], e2 > 0, fabs(e2) > 3 * NGUONG ? 1000 : 300);
      if (fabs(e1) > NGUONG) chay(PV_T, PV_N, QT[0], QT[1], e1 > 0, fabs(e1) > 3 * NGUONG ? 1000 : 300);
    }
    // 3) LCD: gio DS1307 + goc hai khop
    int ng, th, hh, mm, ss; docDS1307(ng, th, hh, mm, ss);
    char d1[17], d2[17];
    sprintf(d1, "%02d/%02d %02d:%02d:%02d", ng, th, hh, mm, ss);
    sprintf(d2, "ng=%+04.0f q=%+04.0f %s", docGoc(POT_NG, 0, 75),
            docGoc(POT_PV, -120, 120), S > S_MIN ? "OK" : "MAY");
    lcd.clear(); lcd.setCursor(0, 0); lcd.print(d1);
    lcd.setCursor(0, 1); lcd.print(d2);
```

```
    Serial.printf("%s S=%.0f e1=%+.0f e2=%+.0f\n", d1, S, e1, e2);
  }
}
```

Code hai tr c gi nguy n khung b^n m t tr c v• th m ba i m: kh i thi n v n tr^ v' c€p (¢, £) kœm chušn h‡a b c nh^y 180¥; nh nh tinh ch nh x•t e2 c a kh p nghi ng tr c r"i m i t i e1 c a kh p ph ng v ; m•i kh p c‡ h•m ch y motor ri ng v i c€p c ng t„c h•nh tr nh c a m nh. Tr ng th i kh‡a c u H sau m•i l†nh nh tr c v t t h›m gi nguy n v tr hai kh p.

CH NG 4. TI'N ° TH±C HI N V K' HO«CH TU²N N Y

4.1. K‡t qu^ †t „…c trong tu¡n

Ho•n th•nh c•i Arduino IDE kœm l´i esp32 v• n p th‹ th•nh c ng l n ESP32 DevKit b^n hai tr c.

Ho•n th•nh h•n hai c u H, b n k nh opto, kh i ngu"n v• bus I¡C chung cho DS1307 v i LCD.

Ho•n th•nh vi t v• n p code hybrid hai tr c: •ng th t nghi ng tr c ph ng v sau, hi n th gi v• hai g‡c kh p l n LCD.

4.2. C˜c c§ng viœc c¡n th‰c hiœn ti‡p

Ph i h…p v i " n m t tr c: chia i c ng vi†c lŠp tr nh, m•i " n l•m m t n‹a ° n‹a c a " n hai tr c g"m "ng b DS1307, b^ng thi n v n hai kh p, hi n th LCD v• an to•n c ng t„c h•nh tr nh; n‹a c a " n m t tr c g"m "c b n k nh quang trŽ v• bi n trŽ h"i ti p, thuŠt to n hybrid v• ph t l†nh c u H; hai b n n p th‹ chung tr n m ch.

Ch y ngo•i tr i tr"n ng•y n„ng, i chi u log hai g‡c kh p v i " th m ph'ng Ž quy n nghi n c u.

Tinh ch nh ng ng e1/e2 v• d•i b c ch y gi^m s l n khŽi ng hai motor.

B, sung ch gi‡ l n: "c l ch v• a t*f*m pin v' v tr ngh n–m ngang.

T I LI U THAM KH•O

[1] I. Reda and A. Andreas, µSolar Position Algorithm for Solar Radiation Applications,¶ NREL/TP-560-34302, 2008. https://docs.nrel.gov/docs/fy08osti/34302.pdf

[2] I. Reda and A. Andreas, µSolar position algorithm for solar radiation applications,¶ Solar Energy, vol. 76, no. 5, pp. 577-589, 2004.

[3] J. A. Duffie and W. A. Beckman, Solar Engineering of Thermal Processes, 4th ed., John Wiley & Sons, 2013.

[4] R. F. Fuentes-Morales et al., µControl algorithms applied to active solar tracking systems: A review,¶ Solar Energy, vol. 212, pp. 203-219, 2020. https://doi.org/10.1016/j.solener.2020.10.071

[5] H. Mousazadeh et al., µA review of principle and sun-tracking methods for maximizing solar systems output,¶ Renewable and Sustainable Energy Reviews, vol. 13, no. 8, pp. 1800-1818, 2009.

[6] A. Awasthi et al., µReview on sun tracking technology in solar PV system,¶ Energy Reports, vol. 6, pp. 392-405, 2020.

[7] A. R. Amelia et al., µTechnologies of solar tracking systems: A review,¶ IOP Conference Series: Materials Science and Engineering, vol. 767, 012052, 2020.

[8] Espressif Systems, ESP32 Series Datasheet. https://www.espressif.com/sites/default/files/documentation/esp32_datasheet_en.pdf

[9] Maxim Integrated, DS1307 64 x 8, Serial I2C Real-Time Clock, Datasheet. https://www.analog.com/media/en/technical-documentation/data-sheets/DS1307.pdf

[10] A. El Hammoumi et al., µA simple and low-cost active dual-axis solar tracker,¶ Energy Science & Engineering, vol. 6, no. 5, pp. 607-620, 2018. https://doi.org/10.1002/ese3.236

[11] J.-M. Wang and C.-L. Lu, µDesign and Implementation of a Sun Tracker with a Dual-Axis Single Motor for an Optical Sensor-Based Photovoltaic System,¶ Sensors, vol. 13, no. 3, pp. 3157-3168, 2013. https://doi.org/10.3390/s130303157